Customer Segmentation Analysis

K-Means clustering | Fresher-level analytics assignment

Dataset: 2,627 customer records, 10 source fields. **Approach:** exploratory K-Means clustering with three segments.

1. Executive summary

The dataset was cleaned and segmented into three customer groups using demographic and stated spending-score attributes. The selected model produced a silhouette score of 0.196, the highest among the tested values k=3, 4, 5 and 6. This is a modest score, indicating that the groups overlap; results should be used as an initial marketing hypothesis rather than a definitive customer classification.

2. Segment profiles

Segment	Customers	Avg. age	Avg. family size	Avg. work exp.	High spend share	Most common professior
S1	1,345	54.6	2.4	0.9	23.6%	Artist
S2	788	28.6	3.9	1.3	3.2%	Healthcare
S3	494	37.8	2.3	8.2	8.9%	Artist

Segment 1 – Mature, higher-spend potential: 1,345 customers (51.2% of the dataset). Spending mix: 23.6% High, 36.3% Average and 40.1% Low. Married share: 86.2%; graduates: 75.8%.

Segment 2 – Younger, low-spend: 788 customers (30.0% of the dataset). Spending mix: 3.2% High, 4.8% Average and 92.0% Low. Married share: 15.7%; graduates: 29.3%.

Segment 3 – Experienced, low-spend: 494 customers (18.8% of the dataset). Spending mix: 8.9% High, 20.0% Average and 71.1% Low. Married share: 48.0%; graduates: 71.3%.

3. Marketing recommendations

Strategy 1 – VIP retention for Segment 1: Create a loyalty tier for customers with High spending scores. Offer early access, priority support and exclusive bundles. Measure repeat purchase rate, revenue per customer and retention.

Strategy 2 – Personalised growth within Segment 1: For customers with Average spending scores, test relevant cross-sell bundles and limited-time upgrades based on product interest. Measure conversion rate and average basket value.

Strategy 3 – Affordable activation for Segments 2 and 3: These groups are predominantly Low spending. Test entry-level bundles, useful onboarding content and small targeted incentives. Track activation, incremental sales and campaign margin; avoid blanket discounts.

4. Data quality and methodology

The source contains 526 missing cells: Ever_Married: 50, Graduated: 24, Profession: 38, Work_Experience: 269, Family_Size: 113, Var_1: 32. Customer IDs are unique. Numeric missing values were imputed with the median and categorical missing values with the mode for model preparation. Numeric fields were standardised and categorical fields one-hot encoded. Customer ID was excluded from clustering.

5. Important limitation: RFM is not possible with this file

The supplied file has no transaction dates, order IDs or purchase amounts. Therefore Recency, Frequency, Monetary value, average order value and order frequency cannot be computed reliably. The analysis uses K-Means on the customer attributes available, including the Spending_Score field. To perform true RFM, add customer-level purchase history with customer ID, order ID, order date and order amount, then calculate a common analysis date, days since last purchase, order count and total spend.

6. Conclusion

The clustering provides three initial customer profiles and actionable hypotheses for differentiated marketing. Segment 1 contains the largest share of High spending-score customers and is the best starting point for VIP retention and personalised upselling. Before deploying campaigns, validate these profiles using actual transaction history and campaign response data.